

SATYAJIT NAYAK

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PROFESSIONAL SUMMARY

Electronics & Communication Engineering graduate (VIT, 2025) specializing in Applied AI/ML, Computer Vision, and Generative AI systems. Proven track record building end-to-end production ML pipelines—from multimodal RAG architectures and deep learning explainability (Grad-CAM, Bayesian uncertainty) to high-throughput LLM gateway routing, Redis semantic caching, and containerized microservices.

TECHNICAL SKILLS

Languages & Frameworks: Python, Java, C++, SQL, FastAPI, Flask, Streamlit, HTML/CSS, REST APIs

Machine Learning & AI: PyTorch, TensorFlow, Keras, OpenCV, Scikit-Learn, Sentence-Transformers, Qdrant, LangChain, EasyOCR, Hugging Face

Cloud, DevOps & Systems: AWS (EC2, S3), Docker, Docker Compose, Redis, Git, GitHub Actions, Linux, Postman, ReportLab

WORK EXPERIENCE

Software Development Intern | *Hindustan Aeronautics Limited (HAL)* June 2024 – Aug 2024

- Built real-time preprocessing and data analytics pipelines handling 15,000+ telemetry data points/sec for predictive maintenance.
- Engineered data validation and anomaly filtering algorithms, cutting downstream diagnostic ingestion latency by 35%.
- Authored component architecture and API specifications ensuring seamless hardware-software integration across embedded systems.

Artificial Intelligence Intern | *USRYA* Sept 2023 – Dec 2023

- Developed modular RESTful inference microservices in FastAPI/Flask connecting deep learning models to interactive client applications.
- Profiled execution bottlenecks and optimized tensor payloads, reducing p95 endpoint latency by 42% (from 380ms to 220ms).
- Implemented automated integration tests and error-handling middleware, achieving 99.5% API uptime during staging trials.

KEY PROJECTS

Brain Tumor MRI Classifier with Explainability & Uncertainty | [\[GitHub\]](#) 2026

- Benchmarked custom CNN baseline against fine-tuned ResNet50 with 5-fold CV on 7,000+ multi-class scans (glioma, meningioma, pituitary, normal).
- Prioritized clinical False Negative Rate (0.00% FNR) with inverse-frequency class weighting to eliminate missed tumor diagnoses.
- Implemented universal Grad-CAM layer resolution and Monte Carlo Dropout (N=20) for Bayesian epistemic uncertainty quantification.
- Built a clinical PDF report generator (ReportLab), FastAPI backend, Streamlit console, and automated CI/CD test workflows.

Multimodal Technical Datasheet RAG System | [\[GitHub\]](#) 2026

- Developed layout-aware extraction pipelines segmenting datasheets into distinct text, table (pdfplumber), and diagram (OpenCV/EasyOCR) vector spaces.
- Implemented multi-collection hybrid retrieval in Qdrant with Cross-Encoder reranking (ms-marco-MiniLM-L-6-v2), boosting precision by 28% over text-only baselines.
- Engineered confidence threshold guards to prevent LLM hallucinations and containerized the architecture with Docker Compose.

Enterprise LLM Gateway & Dynamic Routing Engine | [\[GitHub\]](#) 2026

- Built a high-throughput unified API gateway with dynamic routing, automatic failover, and load balancing across multi-provider LLMs.
- Integrated semantic response caching in Redis to eliminate duplicate upstream LLM calls, cutting token costs by 60% with <15ms cache hits.
- Implemented token-bucket rate limiting, streaming response aggregation, and real-time cost telemetry logging.

EDUCATION

Vellore Institute of Technology (VIT) | *B.Tech in Electronics and Communication Engineering* 2021 – 2025 **SAI International School** | *Class XII (CBSE)* 2018 – 2020

CERTIFICATIONS & LEADERSHIP

Certifications: AWS Certified Solutions Architect – Associate | AWS Cloud Practitioner | Oracle OCI 2025 AI Associate

Leadership: Tech Team Lead, Youth India Foundation (led 20+ developers; scaled to 15k+ users) | Sponsorship Team Lead, VITOPIA (VIT-AP)